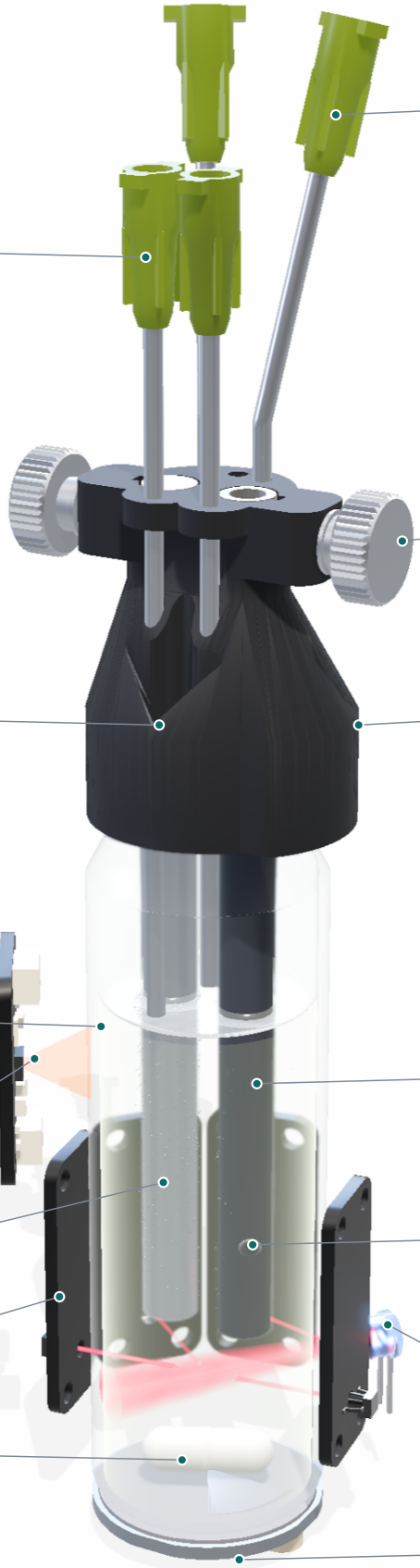
Aseptic electroPioreactor AEP0.2

Vial components



Gas outlets
Two bent needles vent spent gas from above the liquid.

Media In and Media Out needles
Fresh medium is pumped in. The Media Out tip sets the 30 ml level and removes culture as the Pioreactor doses.

Knurled thumb screws
Clamp the power leads to the electrodes: red to the anode, black to the cathode.

Inoculation port
A syringe goes straight through the self-sealing silicone septum.

One-piece vial cap
3D-printed in PC-CF. One silicone septum seals the vial, both electrodes and every port, and the cap sets the electrode depth.

Culture medium, 30 ml
Hydrogen-oxidising bacteria fix CO_2 using the H_2 and O_2 made in the culture.

Precision temperature sensor
An MLX90632 infrared thermometer reads the culture through the glass, without contact.

Anode: hollow MMO-coated titanium
Electrolysis makes O_2 on its surface. CO_2 is fed down its bore.

Cathode: 316 stainless steel
Electrolysis makes fine H_2 bubbles here, the bacteria's energy source.

CO_2 bubbles
Rise from the anode's open base, supplying carbon and sweeping O_2 bubbles off the anode.

Photodiodes (Pioreactor XR)
Read scattered infrared light at 45°, 90° and 135°, plus a reference, to track growth as optical density.

IR LED and light path
Infrared light crosses the culture below the electrodes (drawn in red; invisible in life).

Magnetic stir flea
Keeps the culture mixed and the gases dissolving.

Heater PCB
An aluminium-core heater under the vial holds the set temperature.

In-culture electrolysis makes hydrogen and oxygen only as they are needed, so hydrogen-oxidising bacteria can turn CO_2 into biomass without stored hydrogen. The Pioreactor runs dosing, stirring, heating, optical density, electrolysis and CO_2 sparging automatically.